

Business Continuity by Design

Two numbers decide the architecture. Everything else follows from them.

RTO and RPO are **business decisions**. The architecture is the cheapest thing that meets them.

The continuum, drawn once

01

Backup and restore

Data copied off-site.
Nothing else runs.

02

Pilot light

Data replicating.
Compute defined but
switched off.

03

Warm standby

A scaled-down copy,
running and able to
serve.

04

Multi-site

Full capacity in both
places, both taking
traffic.

ON THE EXAM

Cost rises left to right. RTO and RPO fall. Every question in this module is a position on this line.

The four strategies

STRATEGY	RT0 / RPO	WHAT IS RUNNING
Backup and restore	Hours to a day / hours	Nothing but stored backups
Pilot light	Tens of minutes / minutes	Data replicating; compute off
Warm standby	Minutes / seconds	A scaled-down live copy
Multi-site active-active	Near zero / near zero	Full capacity, both serving

ON THE EXAM

The line between pilot light and warm standby is whether anything is serving requests. Not whether resources exist.

Pilot light or warm standby?

PILOT LIGHT

Ready to start, not running

- Database replicating continuously
- Compute defined but switched off
- You pay for data, not servers
- Recovery means provisioning first

WARM STANDBY

Running, just small

- A scaled-down copy already serving
- Failover means scaling up
- Costs more, every hour
- It works, because it is running

THE TIEBREAKER

Warm standby buys a second benefit: a running copy proves itself continuously. A pilot light is a hypothesis until tested.

AWS Elastic Disaster Recovery

- ◆ **Continuous block-level replication** from on-premises, other clouds or EC2 into a low-cost staging area.
- ◆ **RPO in seconds, RTO typically 5 to 20 minutes.** Pilot-light economics with warm-standby recovery.
- ◆ **Drill instances** test recovery without touching source servers or interrupting replication.
- ◆ **It recovers servers; it does not fail over traffic.** Route 53 or Global Accelerator moves the users.

ON THE EXAM

DRS is the answer for server-based workloads nobody may re-architect. It converts machines to run natively on AWS.

Replicating the data

- ◆ **Aurora Global Database:** typically under a second of lag, cross-Region failover in about a minute.
- ◆ **DynamoDB global tables:** multi-active, so there is no failover step at all.
- ◆ **RDS cross-Region read replicas** for engines with no global option, promoted manually.
- ◆ **S3 Cross-Region Replication**, with Replication Time Control giving a 15-minute SLA.

ON THE EXAM

Your RPO is whatever your slowest replicating datastore is. One nightly snapshot sets it for the whole system.

AWS Backup, and surviving an attacker

- ◆ **Backup plans** apply schedules and lifecycle by tag, across accounts and Regions, from a delegated administrator.
- ◆ **Vault Lock governance mode can be overridden** by a privileged user. Compliance mode cannot, by anyone.
- ◆ **Logically air-gapped vaults** are locked in compliance mode and shareable via RAM, so another account restores directly.
- ◆ **Restore testing** runs scheduled restores, turning recoverability from an assumption into a measurement.

ON THE EXAM

Sharing an air-gapped vault removes the copy step before restoring, which is exactly where recovery time is usually lost.

An untested disaster recovery plan is not a plan. It is a **hypothesis with a budget.**

Reading the scenario

THE SCENARIO SAYS

Hours of downtime acceptable, cost matters most

Tens of minutes, minimal data loss, cost-sensitive

Minutes, and it must be proven to work

On-premises servers, RPO seconds, no re-architecture

Backups must survive a compromised admin

The RTO has never been measured

IT IS POINTING AT

Backup and restore

Pilot light

Warm standby

Elastic Disaster Recovery

Air-gapped vault, compliance mode

Run a drill first

ON THE EXAM

Find the two numbers, then pick the cheapest strategy that meets both. Over-delivering on RTO is a design error.

Meridian Freight's order system must recover within two hours of a Region failure, losing no more than five minutes of data. The business has been explicit that it will not fund a second full production environment.

The system is Aurora PostgreSQL behind an Auto Scaling group and an Application Load Balancer.

What should the solutions architect design?

- ☐ A Backup and restore, with hourly snapshots copied to the second Region
- ☐ B Pilot light: an Aurora Global Database secondary, with launch templates and an ASG at zero desired capacity
- ☐ C Warm standby with a scaled-down ASG running continuously
- ☐ D Multi-site active-active across both Regions

◆ PAUSE AND DECIDE

What should the solutions architect design?

A Backup and restore

Hourly snapshots give an RPO of up to an hour against a five-minute requirement. It fails on data loss, not on time.

C Warm standby

Meets both numbers, and costs more every hour to deliver an RTO of minutes when two hours was asked for.

B Pilot light with Global Database

Sub-second replication meets the RPO, and provisioning compute fits comfortably inside two hours. Nothing idle is paid for.

D Multi-site active-active

The most expensive option available, and it is precisely the second full environment the business ruled out.

THE SIGNAL

Two numbers, and only one strategy meets both at the lowest cost. C and D fail on money rather than capability.

A company must provide disaster recovery for 340 on-premises servers running a mix of Windows and Linux, including several vendor applications nobody may modify. The requirement is an RPO of seconds and an RTO under thirty minutes. There is no appetite to re-architect anything before the deadline.

What should the solutions architect recommend?

- ☐ A AWS Elastic Disaster Recovery replicating into a staging area in AWS
- ☐ B AWS Backup taking hourly backups of the on-premises servers
- ☐ C Rebuild the servers as AMIs, kept updated with EC2 Image Builder
- ☐ D AWS Application Migration Service to move them to AWS permanently

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

Elastic Disaster Recovery

Continuous block replication gives an RPO of seconds, and recovery instances launch in five to twenty minutes. No application changes at all.
- B

Hourly AWS Backup

An RPO of up to an hour against a requirement of seconds. It is the wrong order of magnitude.
- C

AMIs plus Image Builder

An image captures a machine at build time, so the RPO is however old the image is. It also means rebuilding 340 servers.
- D

Application Migration Service

Its close cousin, and it migrates permanently. The requirement is disaster recovery, with production staying on-premises.

THE SIGNAL

MGN migrates; DRS protects. They share technology and answer different questions, which the exam tests directly.

A documented DR plan claims a four-hour RTO based on restoring snapshots into a second Region. It has never been executed. A tabletop review found that the CloudFormation templates reference a KMS key and an IAM role that exist only in the primary Region. The board has asked whether the four-hour figure is credible.

What should the solutions architect recommend FIRST?

- ☐ A Upgrade to warm standby to reduce the RTO
- ☐ B Run a full recovery drill, measure the actual RTO, and fix what it exposes
- ☐ C Create the missing KMS key and IAM role, then declare the plan complete
- ☐ D Increase snapshot frequency to improve the RPO

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Upgrade to warm standby
Spends money to improve a number nobody has measured. The current design might meet four hours once repaired.
- B

 Run a drill and measure
The tabletop found two gaps without executing anything. A real drill finds the rest, and produces a measured figure rather than a claim.
- C

 Fix the two known gaps
Fixes what one review happened to catch, and assumes it caught everything. A tabletop cannot find a slow restore.
- D

 More frequent snapshots
Improves the RPO, which nobody questioned. The board asked about recovery time.

THE SIGNAL

The word FIRST matters. You cannot improve a number you have never measured, or trust one you have never tested.

A company has implemented DRS and successfully tested launching recovery instances in the secondary Region. During a simulated Region failure the recovery instances came up in twelve minutes, but customers continued to receive errors for another forty minutes. Nothing was wrong with the recovery instances themselves.

What was missing?

- ☐ A DRS replication had fallen behind and needed to catch up
- ☐ B Automated traffic failover: DRS launches instances but does not redirect users
- ☐ C The recovery instances needed larger instance types
- ☐ D AWS Backup should have been used instead of DRS

◆ PAUSE AND DECIDE

What was missing?

- A

 Replication lag
Would affect how current the recovered data is, not how long users see errors. The instances launched in twelve minutes.
- B

 Traffic failover was never automated
DRS recovers servers and stops there. Something must move users to them, and that is Route 53 or Global Accelerator.
- C

 Bigger instances
A performance answer to an availability symptom. The scenario says nothing was wrong with the instances.
- D

 Use AWS Backup
Would make recovery slower rather than faster. DRS was the right service; the gap is elsewhere.

THE SIGNAL

Twelve minutes to recover and fifty-two before users are served is the signature of a plan that stops at the servers.

After a ransomware incident at a competitor, a bank asks whether an attacker who compromised an administrator account could destroy both production data and its backups. Backups are currently written to an AWS Backup vault in the same account. The recovery team must be able to restore quickly even if that account is unusable.

What should the solutions architect implement?

- ☐ A Enable versioning on the backup vault
- ☐ B Copy backups to a logically air-gapped vault, shared with a recovery account through AWS RAM
- ☐ C Apply Vault Lock in governance mode to the existing vault
- ☐ D Increase backup frequency and retention

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A

 Versioning
Not a backup vault feature, and an administrator who can delete the vault is not stopped by keeping more copies inside it.
- B

 Air-gapped vault, shared via RAM
Locked in compliance mode so nobody can delete it, and shareable so a separate account restores directly with no copy step.
- C

 Vault Lock, governance mode
The sharpest miss. Governance mode is designed to be overridden by a sufficiently privileged user, which is exactly the threat.
- D

 More backups
More copies in the same account the attacker controls. Frequency was never the gap.

THE SIGNAL

Governance versus compliance mode appears again, exactly as with S3 Object Lock in module ten. The threat model decides it.

A platform replicates Aurora with a Global Database, S3 with Cross-Region Replication and Replication Time Control, and DynamoDB with global tables. A configuration store on Amazon EFS is backed up nightly to the second Region.

The team documents an RPO of one second and is confident it is achievable.

What should the solutions architect tell them?

- ☐ A The documented RPO is correct; all three databases replicate in under a second
- ☐ B The real RPO is up to 24 hours, set by the nightly EFS backup
- ☐ C The real RPO is 15 minutes, set by S3 Replication Time Control
- ☐ D RPO cannot be determined without measuring replication lag

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

- A

 One second is correct
True of the three databases and not of the system. The EFS configuration store is part of the workload.
- B

 24 hours, set by the EFS backup
A system's RPO is its slowest replicating component. A nightly copy means up to a day of configuration loss.
- C

 15 minutes, from RTC
Right reasoning, wrong component. RTC is the slowest of the three replicated stores; the nightly backup is far slower.
- D

 Cannot be determined
It can. A nightly schedule bounds the worst case at 24 hours without measuring anything.

THE SIGNAL

RPO is a property of the whole system, not of its best component. The forgotten datastore usually sets it.

What to carry out of this module

- ◆ **Find the two numbers, then pick the cheapest strategy meeting both.** Over-delivering on RTO is a design error.
- ◆ **Pilot light versus warm standby is whether anything is serving.** Not whether resources exist.
- ◆ **DRS recovers servers, not traffic.** Route 53 or Global Accelerator moves the users.
- ◆ **Your RPO is your slowest replicating datastore.** One nightly backup sets it for everything.

ON THE EXAM

An untested plan is a hypothesis. When a claimed RTO has never been measured, running the drill is almost always step one.